

Mining Quantitative Associations

- ❑ Mining associations with numerical attributes
 - ❑ E.g.: Numerical attributes: **age** and **salary**
- ❑ Methods
 - ❑ **Static discretization** based on predefined concept hierarchies
 - ❑ Discretization on each dimension with hierarchy
 - ❑ age: {0-10, 10-20, ..., 90-100} $\rightarrow$ {young, mid-aged, old}
 - ❑ **Dynamic discretization** based on data distribution
 - ❑ **Clustering**: Distance-based association
 - ❑ First one-dimensional clustering, then association
 - ❑ **Deviation analysis**:
 - ❑ Gender = female $\Rightarrow$ Wage: mean=\$7/hr (overall mean = \$9)

Mining Extraordinary Phenomena in Quantitative Association Mining

- ❑ Mining extraordinary (i.e., interesting) phenomena
 - ❑ E.g.: **Gender = female** $\Rightarrow$ **Wage: mean=\$7/hr** (overall mean = \$9)
 - ❑ **LHS**: a subset of the population
 - ❑ **RHS**: an extraordinary behavior of this subset
- ❑ The rule is accepted only if a statistical test (e.g., Z-test) confirms the inference with high confidence
- ❑ Subrule: Highlights the extraordinary behavior of a subset of the population of the super rule
 - ❑ E.g.: **(Gender = female) ^ (South = yes) $\Rightarrow$ mean wage = \$6.3/hr**
- ❑ Rule condition can be categorical or numerical (quantitative rules)
 - ❑ E.g.: **Education in [14-18] (yrs) $\Rightarrow$ mean wage = \$11.64/hr**

Efficient methods have been developed for mining such rules (e.g., Aumann and Lindell@KDD'99)